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Economic Decisions,
Engineering Costs,
and Cost Estimating

Newman et. al, Oxford Press

Alternative Futures

- Choosing the right vehicle is a complex problem that requires more information than price and fuel mileage.
- In 2016, vehicle lineup included the following sedans:

Model	Technology	Litres per 100 km	MSRP
Nissan Leaf	Electric	2.4	\$32,700
Chevrolet Volt	Electric	3.8	\$40,000
Toyota Prius	Hybrid	4.7	\$26,000
Honda Civic	Hybrid	5.8	\$16,000
BMW X5 xDrive35d	Diesel	6.2	\$87,000
Chevrolet Cruz	Diesel	5.1	\$24,670
Chevrolet Cruze Eco	Gasoline	5.6	\$16,175
Toyota Yaris	Gasoline	4.6	\$16,365

- Future drivers will have to use engineering economic analysis to find the correct choices.

Learning Objectives

- Describe the *economic decision-making process*
- Choose economic *criteria* for different problems
- Describe common ethical issues in economic decisions
- Define various cost concepts and their importance
- Define engineering cost estimating
- Explain the three types of engineering estimates and the difficulties encountered in making them
- Use common mathematical cost estimating models
- Discuss the effect of the *learning curve* on cost estimates
- Draw *cash flow diagrams* to show project costs/benefits

A Sea of Problems

- Problems can be classified by levels of difficulty
 - **Simple** (do not require much time or effort)
 - If we use three crates of an item a week, how many crates should we buy at a time?
 - **Intermediate** (primarily economic)
 - Should we buy a low-cost press requiring three operators, or a more expensive one requiring only two operators?
 - **Complex** (mix of economic, political, and social/ethical)
 - The annual budget of a corporation is an allocation of resources, and all projects are evaluated economically. The budget process is also heavily influenced by non-economic forces such as political or national concerns, individual concerns, and other corporation-wide impacts.

The Role of Engineering Economic Analysis

- Engineering economic analysis is most suitable for intermediate problems and the economic aspects of complex problems
- Such problems should:
 - be *important enough* to justify serious thought and effort;
 - be not easily worked out in one's head; and
 - contain *economic* aspects that are important in making the decision.

The Decision-Making Process

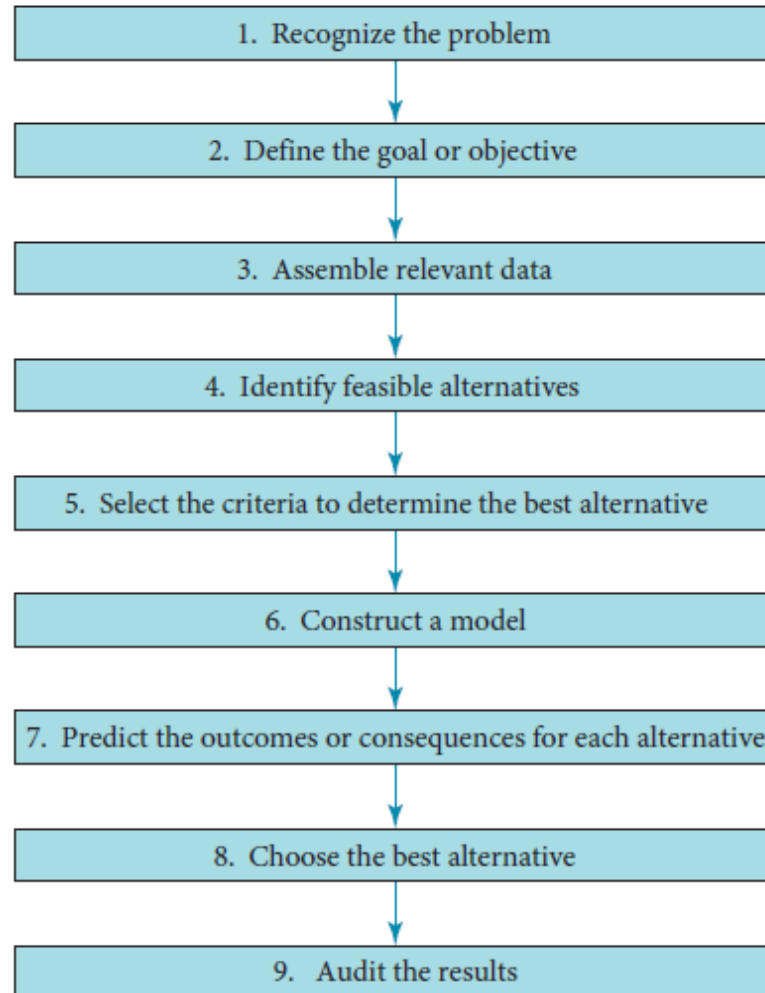


FIGURE 1-1 One possible flow chart of the decision-making process.

The Decision-Making Process, cont'd

1) Recognize the Problem

- Once we are aware of the problem, we solve it as best we can

2) Define the goal or objective

- The goal can be a grand, overall goal of a person or a firm
- An objective may be narrow and specific

The Decision-Making Process, cont'd

3) Assemble Relevant Data

- Obtain the information through market research or other data gathering
- An important source of data is a firm's own accounting system
- The time horizon of the problem must be assembled
- Financial consequences (costs and benefits):
 - Market consequences
 - Extra-market consequences
 - Intangible consequences

The Decision-Making Process, cont'd

4) Identify Feasible Alternatives

- Unless the best alternative is considered, the result will always be less than ideal
- Two types of alternatives are sometimes ignored:
 - “Do-nothing” alternative
 - Unglamorous alternatives
- Ensure that all alternatives have been listed
- Impractical alternatives may lead to a better possibility
 - Brainstorming

The Decision-Making Process, cont'd

5) Select the Criteria for Determining the Best Alternative

- We must define what we mean by “best”
- There must be at least one criterion, or a set of criteria, to evaluate which alternative is best
- "Maximize profit" is the criterion normally selected in engineering decision-making

The Decision-Making Process, cont'd

6) Construct a Model

- The objective, relevant data, feasible alternatives, and selection standards must be merged
 - Model building
- In economic decision-making, the model is usually mathematical

The Decision-Making Process, cont'd

- 7) Predict the Outcomes for Each Alternative
 - The model and data are used to predict the outcomes of each feasible alternative
 - To choose the best alternative, the outcomes for each alternative must be stated in a *comparable* way
 - The consequences of alternatives are usually evaluated in monetary terms
 - A common mistake for longer-term problems is to assume that the current situation will be unchanged if the do-nothing alternative is chosen

The Decision-Making Process, cont'd

8) Choose the Best Alternative

- The right choice is the one that best meets the criteria after we have considered both the numerical and intangible consequences
- Since intangible consequences are left out of the numerical calculations, they should be introduced into the decision-making at this point

The Decision-Making Process, cont'd

9) Audit the Results

- An audit of the results is a comparison of what happened against the predictions
- Audits help keep a project on track
- Audits help future estimates and assumptions
- Audits provide an incentive to give accurate estimates

Decision-Making: Example 1

A company is considering two machines. Both machines provide the same output and benefits. Machine 1 costs \$300,000 to purchase and requires four employees to operate. The costs of maintenance of Machine 1 is \$100,000 and \$50,000 per employee over the machine's lifetime. Machine 2 costs \$500,000 to purchase and requires two employees to operate. The machine is maintenance free and costs \$40,000 per employee over the machine's lifetime. Which machine should the company purchase?

Decision-Making: Example 2

John, an auto mechanic, inherited a sum of \$1,000,000 from his parents. He is making about \$15 an hour and works about 50 hours a week for 52 weeks a year (10 hour overtime per week). He gets two weeks of paid vacation. The overtime wage is at time-and-a-half of the regular wages.

He would like to get an income from his inheritance equivalent to his job so that he can quit his job. He found that he could invest the money safely to get an interest of 5% per year for the rest of his life.

Will John make enough money with his investments to quit his job?

Ethics

- Ethics can be described as distinguishing right and wrong when making decisions
 - Establishing beliefs and moral obligations
 - Defining values and fairness
 - Determine duties and guidelines for conduct
- Ethical decision-making requires an understanding of the “context” of the problem, the possible choices, and the outcomes of each choice

Ethical Dimensions in Engineering Decision-Making

- In Canada, provincial and territorial associations of professional engineers are responsible for the regulation of the practice of engineering
 - Engineers Canada
- For all engineers, difficulties arise when their actions are contrary to these written or internal codes

The Environment We Live In

- The decision maker must ask who incurs the costs for the project and who receives the benefits
- Ethical issues can be particularly difficult because there are often stakeholders with opposing viewpoints
 - E.g., protecting the habitat of an endangered species versus flood-control projects that protect people, animals, and structures

Safety and Cost

- There are trade-offs between safety and cost
 - Is it safe enough?
 - It is “too safe,” therefore too expensive?

Emerging Issues and “Solutions”

- Governments have started to prevent, limit, and expose financial wrongdoing within corporations
- Globalization is an important part of ethical discussions
 - Different countries have different ethical expectations



Importance of Ethics In Engineering and Engineering Economy

- Though engineers and firms try to act ethically, mistakes are made
- It is the engineers' duty to speak out to protect the safety of the public
- Integrity is the foundation for long-term career success



Engineering Decision Making for Current Costs

- If a decision is one where the results are known in a short period of time, it can be determined by:
 - Adding up costs and benefits between alternatives
 - Thereby, one can determine the best alternative

Engineering Decision Making for Current Costs, cont'd

- Example: (Tile for Plant Floor—fixed output)
 - 1000 m²
 - Tile *A* costs:
 - Purchase: \$52/m²; Installation: \$37,000
 - Tile *B* costs:
 - Purchase: \$63/m²; Installation: \$28,000
- Select Tile *A* to minimize cost for fixed output

Engineering Costs

- Fixed, Variable, Marginal, and Average Costs
 - **Fixed:** Constant, unchanging costs
 - **Variable:** Depends on the level of output or activity
 - **Marginal:** Variable cost for one more unit
 - **Average:** Total cost divided by the number of units

Engineering Costs, cont'd

- Fixed, Variable, Marginal, and Average Costs, cont'd
 - **Break-even point:** The level of activity at which the total cost of providing the product, good, or service is equal to the revenue generated
 - **Profit region:** Values of the variable x greater than the break-even point, where total revenue is greater than total costs
 - **Loss region:** Values of the variable x less than the break-even point, where total cost is greater than total revenue

Engineering Costs, cont'd

Engineering Costs: Example 3

- A company operates a summer camp. The following cost data for a 12-week summer camp is as follows:
 - Charge per camper = \$400/week
 - Variable cost per camper = \$220/week
 - Fixed costs = \$240,000 per summer season
 - Capacity per week = 200 campers
- Determine the following:
 - The total number of campers to breakeven for the season
 - The profit if the camp is operated at 90% capacity
 - The additional profit that can be made if a discount of \$100 per week is given for another 10 campers

Engineering Costs, cont'd

- Sunk Costs
 - Money already spent due to a past decision
 - Should be disregarded in engineering economic analysis

Engineering Costs, cont'd

- Opportunity Costs
 - The costs associated with a resource being used for an alternate task
 - Sometimes referred to as “forgone opportunity costs”
 - “An opportunity cost is the benefit that is forgone by engaging a business resource in a chosen activity instead of engaging that same resource in a forgone activity.”

Engineering Costs, cont'd

- Recurring and Non-Recurring Costs
 - **Recurring:** A cost that reoccurs at regular intervals
 - **Non-recurring :** One-of-a-kind cost recurring at irregular intervals

Engineering Costs, cont'd

- Incremental Costs
 - **Incremental:** Cost differences between alternatives

Engineering Costs, cont'd

- Cash Costs versus Book Costs
 - **Cash costs:** require a cash transaction out of one person's pocket into the pocket of someone else
 - Book costs are recorded but are not transactions
 - Do not represent cash flows, thus are not included in engineering economic analysis

Engineering Costs, cont'd

- Life-Cycle Costs
 - Designing products, goods, and services with a full and explicit recognition of the associated costs over their life cycles
 - Two key concepts:
 - The later a design change is made, the higher the cost
 - Decisions made early in the life cycle tend to “lock in” costs that will be incurred later

Engineering Costs, cont'd

- Life-Cycle Costs

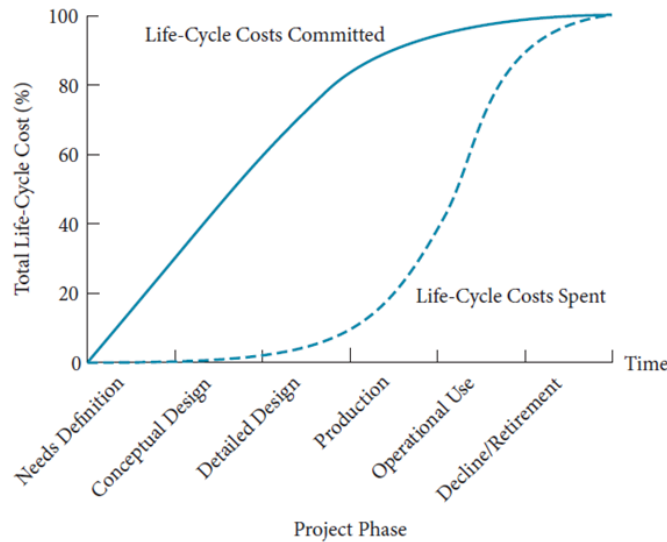


FIGURE 1-4 Cumulative life-cycle costs committed and dollars spent.

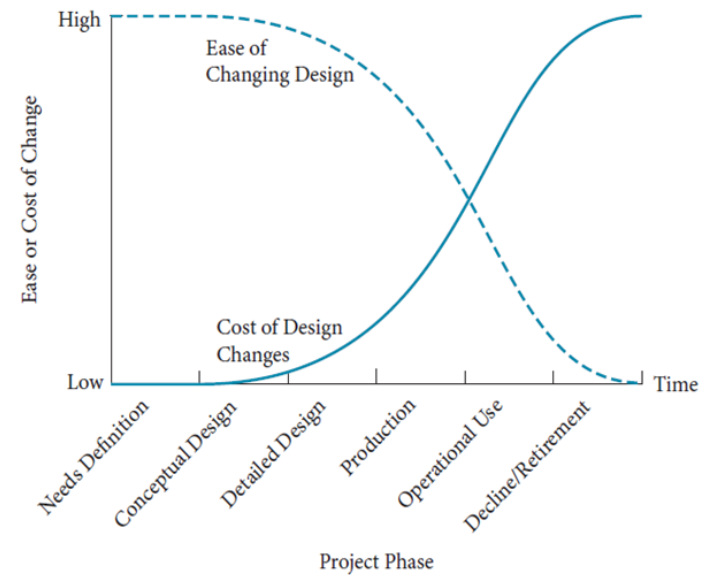


FIGURE 1-5 Life-cycle design change costs and ease of change.

Cost Estimating

- Types of Estimate
 - Rough Estimates
 - Quick/easy, high-level estimates where accuracy varies widely (accuracy of -30% to $+60\%$)
 - Budget Estimates
 - Used for budgeting projects where the accuracy is better than a rough estimate due to the extra effort used to make a determination (accuracy of -15% to $+20\%$)
 - Detailed Estimates
 - Estimates made from detailed designs using quantitative models and vendor quotes. High level of accuracy (accuracy of -3% to $+5\%$)

Cost Estimating, cont'd

- Difficulties in Estimation
 - One-of-a-Kind Estimates
 - First-run projects and projects that have never been done before
 - Time and Effort Available
 - Human resources and time available for making estimates

Estimating Models

- Per-Unit Model
 - Uses a per-unit factor (e.g., cost per square metre)
- Segmenting Model
 - “Divide and conquer” approach
 - Individual and component estimates are added together
- Cost Indexes
 - Historical change in costs as a ratio relationship

$$\frac{\text{Cost at Time A}}{\text{Cost at Time B}} = \frac{\text{Index value at Time A}}{\text{Index value at Time B}}$$

Estimating Models, cont'd

- Power-Sizing Model
 - Used to “scale up” or “scale down” known costs

$$\frac{\text{Cost of A}}{\text{Cost of B}} = \left(\frac{\text{Capacity of A}}{\text{Capacity of B}} \right)^x$$

Where x is the power-sizing component

- $x = 1.0$ shows a linear power sizing component
- $x < 1.0$ shows the “economies” of scale
 - i.e., the more you buy, the cheaper per unit
- $x > 1.0$ shows a “diseconomies” of scale

Estimating Models, cont'd

- Triangulation
 - Approaching the estimate using different sources of data or different quantitative models to confirm the value initially calculated

Estimating Models, cont'd

Cost Indexes: Example 1-7, page 26

- Mariam is interested in estimating the annual labor and material costs for a new project facility. She was able to obtain the following cost data
 - Labor cost index value was at 124 ten years ago and is 188 today
 - Annual labor cost for a similar facility were \$575,500 ten year ago
 - Material cost index value was at 544 three years ago and is 715 today
 - Annual material cost for a similar facility were \$2,455,000 three year ago

Determine the annual cost today for labor and material.

Estimating Models, cont'd

- Improvement and the Learning Curve
 - A percentage or rate at which output is increased due to repetition

$$T_N = T_{initial} \times N^b$$

- T_N = time required for Nth unit of production
- $T_{initial}$ = time required for the first unit of production
- N = number of completed units
- b = learning curve exponent

$$b = \frac{\log(\text{learning curve expressed as a decimal})}{\log 2.0}$$

Estimating Models, cont'd

- The Learning Curve: Example 4
 - In a complex assembly operation, it is found the learning curve rate is 70%. The standard time of 3 minutes per assembly is reached after the 110th unit.
 - Calculate the time required for the very first unit.
 - Calculate the time required for the 200th unit.

Estimating Benefits

- Illustrates the size, sign, and timing of individual cash flows

Timing of Cash Flow	Size of Cash Flow
time 0 (now or today)	positive cash flow of \$100
1 time period from today	negative cash flow of \$100
2 time periods from today	positive cash flow of \$100
3 time periods from today	negative cash flow of \$150
4 time periods from today	negative cash flow of \$150
5 time periods from today	positive cash flow of \$50

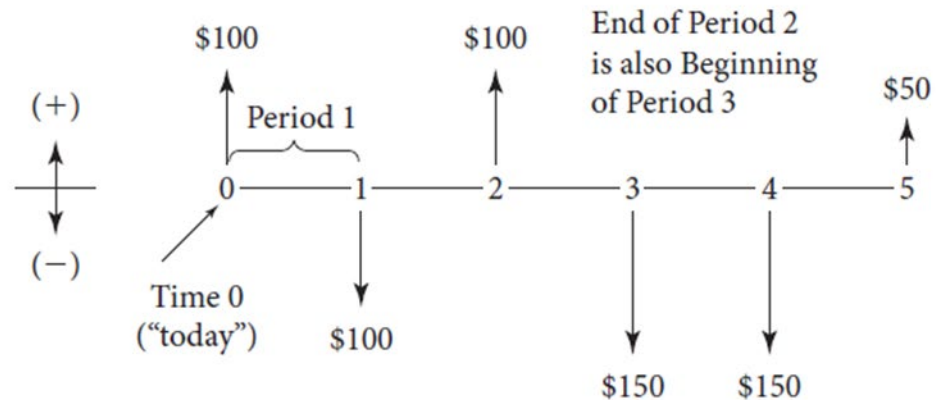


FIGURE 1-7 An example of a cash flow diagram (CFD).

Cash Flow Diagrams

- Economic analysis often requires considering the benefits as well as the costs
- Many of the same methods used to calculate costs can be used to calculate benefits
- Benefits are typically in the future, which sometimes makes them more difficult to estimate

Cash Flow Diagrams

- Categories of Cash Flow
 - **First cost:** Expense of building or of buying and installing
 - **Operations and maintenance (O&M):** Annual expense, such as electricity, labour, and minor repairs
 - **Salvage value:** Receipt at project termination for sale or transfer of the equipment (can be a salvage cost)
 - **Revenues:** Annual receipts due to sale of products or services
 - **Overhaul:** Major capital expenditure that occurs during the life of the asset

Cash Flow Diagrams

- Drawing a Cash Flow Diagram
 - The Cash Flow Diagram (CFD) shows when all cash flows occur
 - Look at Figure 1-7 and the \$100 positive cash flow at the end of Period 2
 - In a CFD, the end of Period N is always the same time as the beginning of Period $N+1$